
1. $\overline{x \vee y} \equiv \bar{x} \wedge \bar{y}$

x	y	$x \vee y$	$\overline{x \vee y}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

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x	y	$\bar{x}$	$\bar{y}$	$\bar{x} \wedge \bar{y}$
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	0

2. $\overline{x \wedge y} \equiv \bar{x} \vee \bar{y}$

x	y	$x \wedge y$	$\overline{x \wedge y}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

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x	y	$\bar{x}$	$\bar{y}$	$\bar{x} \vee \bar{y}$
0	0	1	1	1
0	1	1	0	1
1	0	0	1	1
1	1	0	0	0

3. $\psi_8(x_1, x_2) = x_1 \downarrow x_2 = \overline{x \vee y}$

x_1	x_2	$\psi_8(x_1, x_2)$
0	0	1
0	1	0
1	0	0
1	1	0

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x_1	x_2	$x_1 \downarrow x_2$
0	0	1
0	1	0
1	0	0
1	1	0

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x	y	$\overline{x \vee y}$
0	0	1
0	1	0
1	0	0
1	1	0

4. $\psi_{14}(x_1, x_2) = x_1 \mid x_2 \equiv \overline{x \wedge y}$

x_1	x_2	$\psi_{14}(x_1, x_2)$
0	0	1
0	1	1
1	0	1
1	1	0

=

x_1	x_2	$x_1 \mid x_2$
0	0	1
0	1	1
1	0	1
1	1	0

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x	y	$\overline{x \wedge y}$
0	0	1
0	1	1
1	0	1
1	1	0

5. $f_1(x, y, z) = x \implies (y \vee \bar{z})$

x	y	z	$\bar{z}$	$y \vee \bar{z}$	$x \implies (y \vee \bar{z})$
0	0	0	1	1	1
0	0	1	0	0	1
0	1	0	1	1	1
0	1	1	0	1	1
1	0	0	1	1	1
1	0	1	0	0	0
1	1	0	1	1	1
1	1	1	0	1	1

6. $f_2(x, y, z) = (x \implies y) \vee (y \implies z)$

x	y	z	$x \implies y$	$y \implies z$	$(x \implies y) \vee (y \implies z)$
0	0	0	1	1	1
0	0	1	1	1	1
0	1	0	1	0	1
0	1	1	1	1	1
1	0	0	0	1	1
1	0	1	0	1	1
1	1	0	1	0	1
1	1	1	1	1	1

7. $f_3(x, y, z) = (x \wedge y) \iff (x \implies \bar{z})$

x	y	z	$\bar{z}$	$x \wedge y$	$y \implies \bar{z}$	$(x \wedge y) \iff (y \implies \bar{z})$
0	0	0	1	0	1	0
0	0	1	0	0	1	0
0	1	0	1	0	0	1
0	1	1	0	0	1	0
1	0	0	1	0	1	0
1	0	1	0	0	1	0
1	1	0	1	1	0	0
1	1	1	0	1	1	1
